LAB 05

PH142 Spring 2026

Announcements

- **7/22: Quiz12, Lab05**
- **7/23: Quiz13**
- **7/24: Quiz14**

Data Project Part II is due next Friday. Start working on it early!

Lab05 Lecture Review

Independence of Events

Independent Events

- A and B are independent if knowing whether A occurs <u>does not</u> change the probability of B

Dependent Events

- A and B are dependent if knowing whether A occurs <u>does</u> change the probability of B

Lab05 Lecture Review

Screening Tests: Key Terms

Sensitivity: Probability the test is positive given disease present

- Sensitivity = **P(Test+ | Disease+)**

Specificity: Probability the test is negative given disease absent

- Specificity = **P(Test- | Disease-)**

Sensitivity and specificity are properties of the test. They do not change based on prevalence of disease in the population.

Lab05 Lecture Review

Predictive Values

Positive Predictive Value (PPV): Probability the disease is present given a positive test result

- PPV = **P(Disease+ | Test+)**

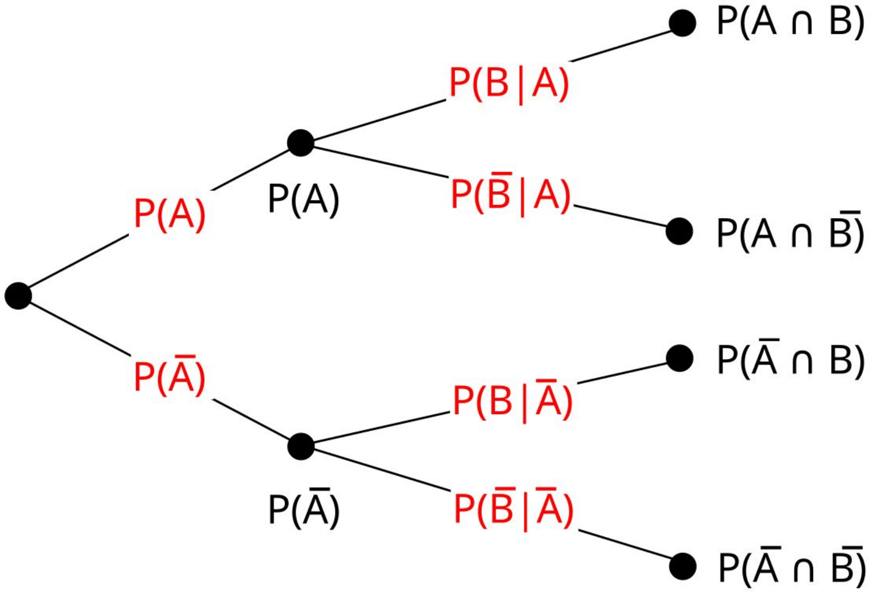
Negative Predictive Value (NPV): Probability the disease is absent given a negative test result

- NPV = **P(Disease- | Test-)**

PPV and NPV change with the prevalence of disease in the population.

Lab05 Lecture Review

Tree Diagrams



Lab05 Lecture Review

The Normal Distribution

The normal distribution is a bell-shaped, symmetric distribution, with notation **X ~ N(μ, σ)**

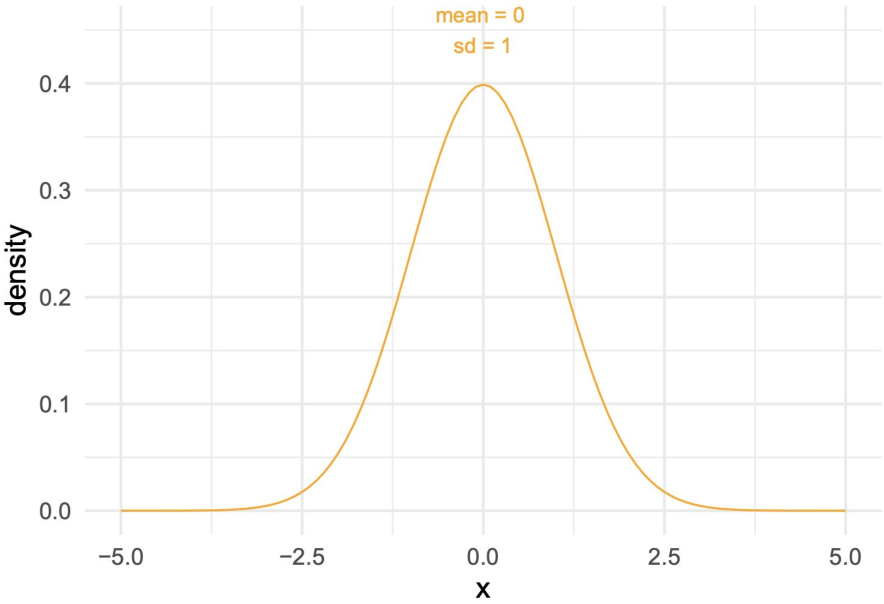
- μ = mean
- σ = standard deviation

R Functions:

- pnorm() → outputs the probability of value x or below
- rnorm() → generates random draws from the distribution

Lab05 Lecture Review

The Normal Distribution



Lab05 Lecture Review

Z-Scores

Z-score formula: $$z = \frac{x - \mu}{\sigma}$$

x = data point
μ = mean
σ = standard deviation

The Z-score tells you how many standard deviations the data point is from the mean.

LAB 05 Walkthrough

Lab Submission

- Follow the directions on the LAB05 file
- Submit using the **Terminal Tab** (next to the console in the bottom left pane)
  - Copy and paste the given line into the terminal
  - Follow prompts (NOTE: the terminal will **not** show your password being typed out!)
- **CHECK IN GRADESCOPE THAT ALL YOUR TESTS PASSED**